Exp 35:Implement a decision tree algorithm for a simple classification problem.

Code:

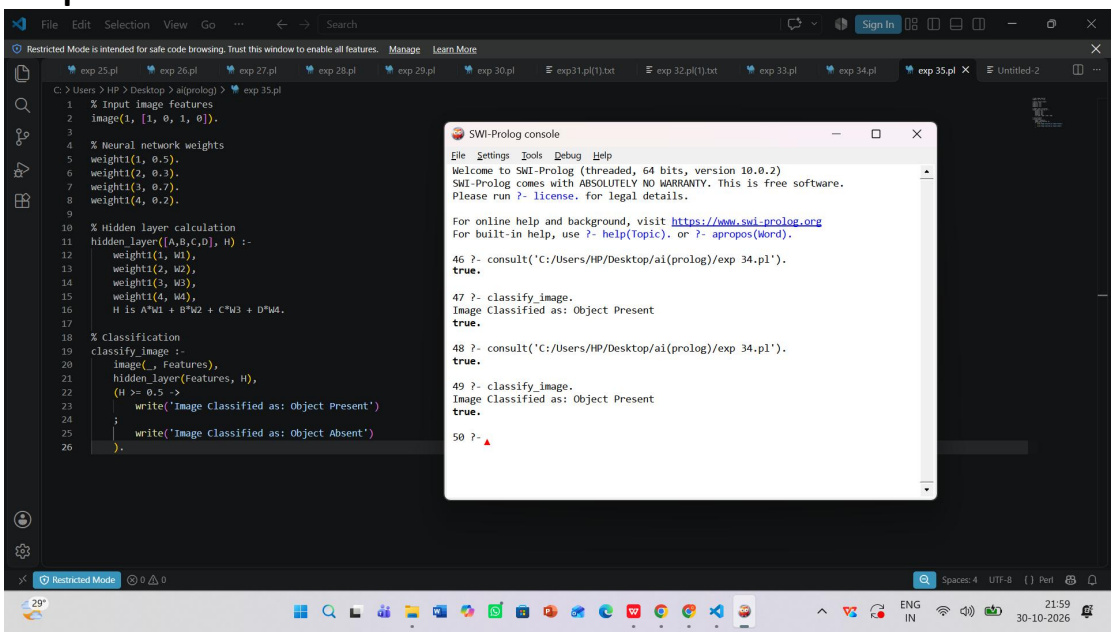
```
% Input image features
image(1, [1, 0, 1, 0]).

% Neural network weights
weight1(1, 0.5).
weight1(2, 0.3).
weight1(3, 0.7).
weight1(4, 0.2).

% Hidden layer calculation
hidden_layer([A,B,C,D], H) :-
   weight1(1, W1),
   weight1(2, W2),
   weight1(3, W3),
   weight1(4, W4),
   H is A*W1 + B*W2 + C*W3 + D*W4.

% Classification
classify_image :-
   image(_, Features),
   hidden_layer(Features, H),
   (H >= 0.5 ->
      write('Image Classified as: Object Present')
   ;
      write('Image Classified as: Object Absent')
   ).
```

Output:


File Edit Selection View Go
Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More
exp 25.pl exp 26.pl exp 27.pl exp 28.pl exp 29.pl exp 30.pl exp31.pl(1).txt exp 32.pl(1).txt exp 33.pl exp 34.pl exp 35.pl Untitled-2
C: > Users > HP > Desktop > ai(prolog) > exp 35.pl
% Input image features
image(1, [1, 0, 1, 0]).
% Neural network weights
weight1(1, 0.5).
weight1(2, 0.3).
weight1(3, 0.7).
weight1(4, 0.2).
% Hidden layer calculation
hidden_layer([A,B,C,D], H) :-
weight1(1, W1),
weight1(2, W2),
weight1(3, W3),
weight1(4, W4),
H is A*W1 + B*W2 + C*W3 + D*W4.
% Classification
classify_image :-
image(_, Features),
hidden_layer(Features, H),
(H >= 0.5 ->
write('Image Classified as: Object Present')
;
write('Image Classified as: Object Absent')
).
SWI-Prolog console
File Settings Tools Debug Help
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
46 ?- consult('C:/Users/HP/Desktop/ai(prolog)/exp 34.pl').
true.
47 ?- classify_image.
Image Classified as: Object Present
true.
48 ?- consult('C:/Users/HP/Desktop/ai(prolog)/exp 34.pl').
true.
49 ?- classify_image.
Image Classified as: Object Present
true.
50 ?-
Restricted Mode
Spaces: 4 UTF-8 Perl
ENG IN
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